

Prince Asumadu

📍 Kumasi - Ghana ✉️ pasumadu6@st.knsut.edu.gh ☎️ 0548 24 70 98 🔗 [portfolio](#) [in eurus08](#) [eurus08](#)

Research Interest

- Scientific Computing
- Machine Learning
- Numerical Analysis
- High-Performance Computing
- Optimization
- Graph Theory

Education

- MPhil** **National Institute of Mathematical Sciences (NIMS-KNUST)** GHANA, Jan 2025 – Present
Scientific Computing and Industrial Modelling
- **Coursework:** Scientific Computing, Advanced Linear Algebra, Machine Learning, HPC ...
 - **Expected completion:** December 2026
- BSc** **Kwame Nkrumah University of Science and Technology (KNUST)** GHANA, Sept 2019 – Nov 2023
Mathematics
- GPA: 3.73/4.0
 - **Coursework:** Linear Algebra, Differential Equations, Functional Analysis, Numerical Analysis, ...

Awards and Scholarships

- SCIM Scholarship** National Institute of Mathematical Sciences 2025 - 2026

Research Experience

- Optimal Task Allocation using the Primal Simplex Method,** Jan 2024 – May 2024
National Institute of Mathematical Sciences, NIMS
- Built an LP model to allocate tasks to employees
 - Modified primal simplex method to solve model.
 - Performed comparative accuracy analysis and benchmarking on algorithms.
- Dynamic Evacuation Plan Model for a Building Complex,** Apr 2023 – Aug 2023
Department of Mathematics, KNUST
Supervisor: Prof. Peter Amoako-Yirenkyi
- Created network graph for case study building complex.
 - Built a dynamic shortest-path model based on network graph.
 - Ran Simulations for different hazardous scenarios.
- National Institute of Mathematical Sciences,** Kumasi
Research Assistant Jan 2024 – Aug 2024
- Performed statistical surveying and code testing for ongoing research at the institute.
 - Conducted individual research project on employee scheduling.

Teaching and Work Experience

Adepts Ghana,

Part-time Database Developer

Kumasi
2024 – Present

- Design and build databases for projects as part of the database team.
- Maintain databases for live projects.

Ghana Computational Biosciences Network,

Python Tutor

Kumasi
Oct 2025 – Nov 2025

- Coordinated a two-month python boot camp for members of the network.

Department of Mathematics (KNUST),

Teaching Assistant

Kumasi
Jan 2024 – Dec 2024

- Tutored students in differential equations and numerical analysis courses.
- Instructed and graded lab sections on openmp and MPI with C.

Ghana Ports and Harbor Authority,

IT Department, Intern

Tema
Oct 2022 – Dec 2022

- Fixed software and hardware issues on staff computers.
- Monitored and set up LAN connections, Wi-Fi, and fibre optics.

Manhyia Hospital,

Research Department, Intern

Kumasi
Oct 2021 – Nov 2021

- Performed daily statistical analysis on out-patient department.
- Conducted individual project on staff satisfaction at the hospital.

Kumasi Academy,

Substitute Teacher

Kumasi
Aug 2020

- Instructed students on elective mathematics in online sessions.

Computational Skills

Programming: • Python (Scikit-learn, PyTorch, Numpy, Pandas, Altair, Seaborn)

• C/C++

• R

• SQL (PostgreSQL)

HPC/Scripting: • MPI

• OpenMP

• CUDA

• $\text{\LaTeX}$

• HTML / CSS

• Shell scripting (Bash)

Others:

• Git / Github

• SPSS

• MATLAB

Training and Conferences

• High Performance Computing for Sustainable Development

Apr 2023

• Deep Learning Indaba

Sep 2023

• DS-Change Workshop: Uniting Academia & Industry

Dec 2023

• Indaba X: Ghana Data Science Summit

Oct 2024